

Module 15: Deadlock Detection & Recovery Strategies

1. Wait-For Graph (WFG) & Recovery Flow

Deadlock Detection (Wait-For Graph) & Recovery Protocols

Wait-For Graph (WFG - Single Instance)

```
graph LR; P1((P1)) --> P2((P2)); P2 --> P3((P3)); P3 --> P4((P4)); P4 --> P1
```

Resource Nodes Collapsed!
Pi → Pj edge indicates Pi is waiting for resource held by Pj.
Cycle detection in WFG runs in $O(V^2)$ via DFS/Tarjan.

Deadlock Recovery Strategies

1. Process Termination

- Abort All Deadlocked Processes: Expensive, but guarantees recovery.
- Abort One-by-One: Abort 1 process, rerun detection. Repeat until free.
- Victim Selection: Lowest CPU time spent, lowest resources held, lowest priority, batch vs interactive.

2. Resource Preemption (Gateway Slide 260 Trace)

- Selecting a Victim: Minimize cost of preempted resources.
- Rollback: Roll back victim process to previous safe checkpoint. (Total rollback = abort and restart; Partial = checkpoint restart).
- Starvation Issue: Same process must NOT be picked as victim repeatedly!
- Solution: Include number of rollbacks in victim selection cost function.

2. Recovery Summary Table

Method	Mechanism	Side Effect & Mitigation
Abort All	Kill all deadlocked processes simultaneously	High computation loss
Abort One-by-One	Kill lowest priority victim repeatedly	Detection rerun overhead
Resource Preemption	Revoke resources and rollback to checkpoint	Starvation risk (mitigated by rollback counter)